Boundary Layer Over a Heated Flat Plate with Surface Radiation

OpenFOAM 13 | Steady Laminar Forced Convection | Grey-Diffuse Surface Radiation

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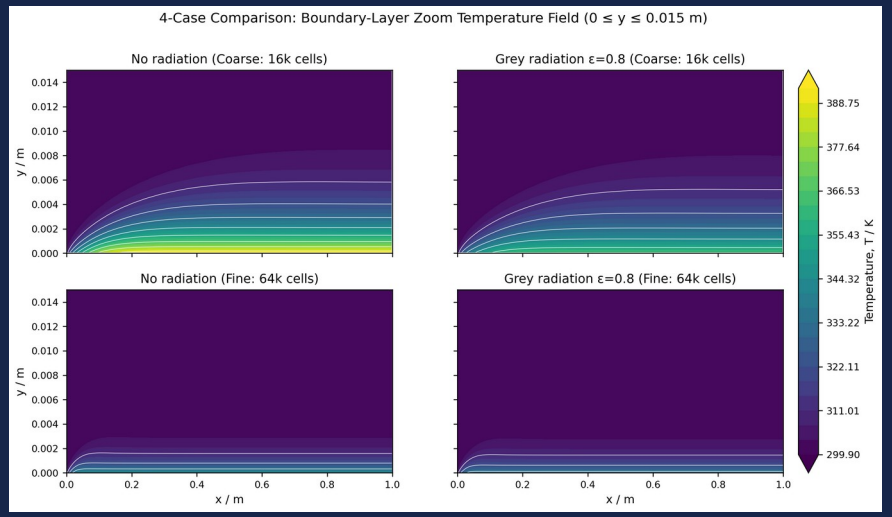


Fig. 1: Near-wall temperature contours across cases.

Problem Definition and Test Matrix

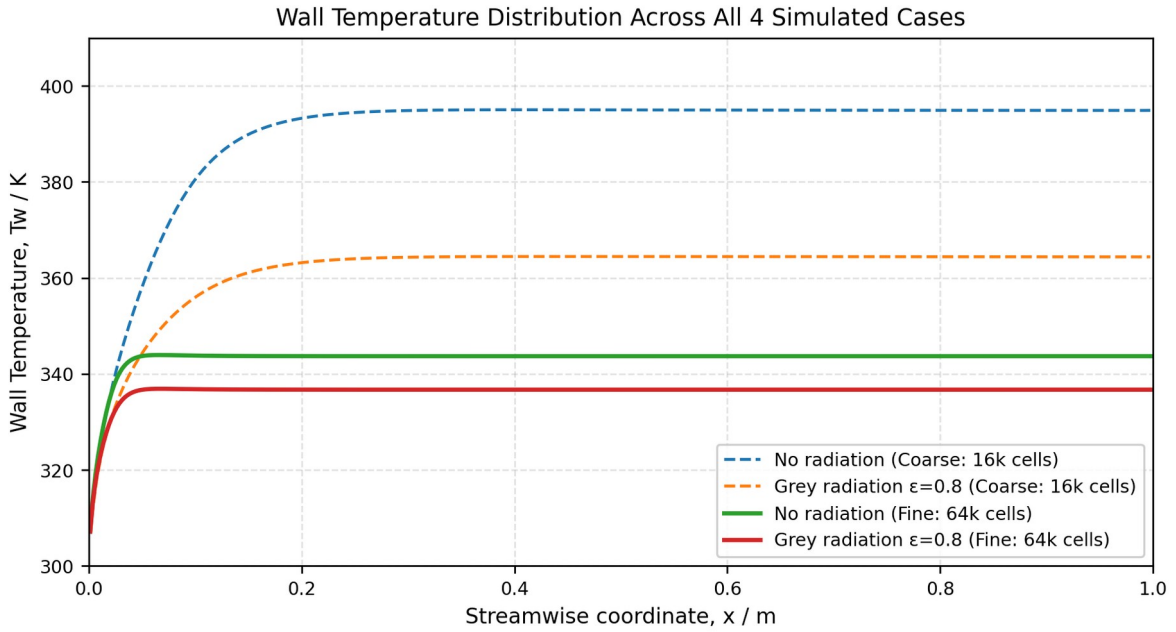


Fig. 2: Streamwise wall temperature distribution along plate.

Core Engineering Question:

How does surface radiation change wall temperature under fixed flux?

- Compares emissivity levels of 0.0 and 0.8 across meshes.
- Evaluates 16,000-cell coarse and 64,000-cell fine structured grids.
- Assumes low-speed, constant-property air flow without buoyancy.

Scope: Low-speed constant-property CFD. Not hypersonic or validated TPS.

Geometry and Input Parameters

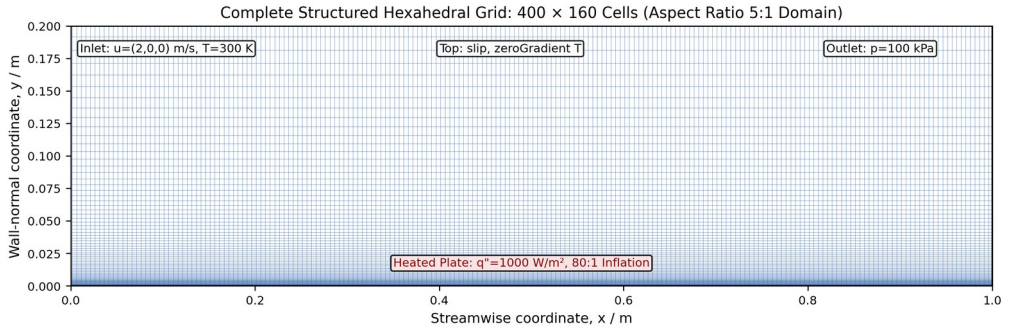


Fig. 3: Computational domain schematic (L = 1.0 m, H = 0.20 m).

Parameter	Value	Unit
Plate length, L	1.0	m
Domain height, H	0.20	m
Inlet velocity, U_inf	2.0	m/s
Inlet temperature, T_inf	300.0	K
Surroundings temperature, T_sur	300.0	K
Reference pressure, p_ref	100.0	kPa
Applied heater flux, q''_heater	1000.0	W/m²
Dynamic viscosity, mu	1.846 x 10^-5	kg/(m s)
Thermal conductivity, k	0.02613	W/(m K)
Specific heat capacity, cp	1005.0	J/(kg K)
Prandtl number, Pr	0.71	—
Density, rho	1.1614	kg/m³

Governing Equations and Physical Assumptions

1. Continuity Equation (Incompressible):

$$\text{div}(U) = 0$$

2. Momentum Conservation (Zero Buoyancy):

$$\rho (U \cdot \nabla) U = -\nabla p + \mu \nabla^2 U$$

3. Thermal Energy Equation:

$$\rho c_p U \cdot \nabla T = \text{div}(k \nabla T)$$

4. Plate Reynolds Number:

$$Re_L = (\rho U_\infty L) / \mu = 1.258 \times 10^5$$

Physical Assumptions & Limitations:

- Steady two-dimensional flow with constant fluid physical properties.
- Buoyancy forces neglected; natural convection effects are completely omitted.
- Re_L supports laminar flow, but ignores physical plate roughness.

Fluid medium is non-participating; no volumetric radiation absorption occurs.

Boundary Conditions

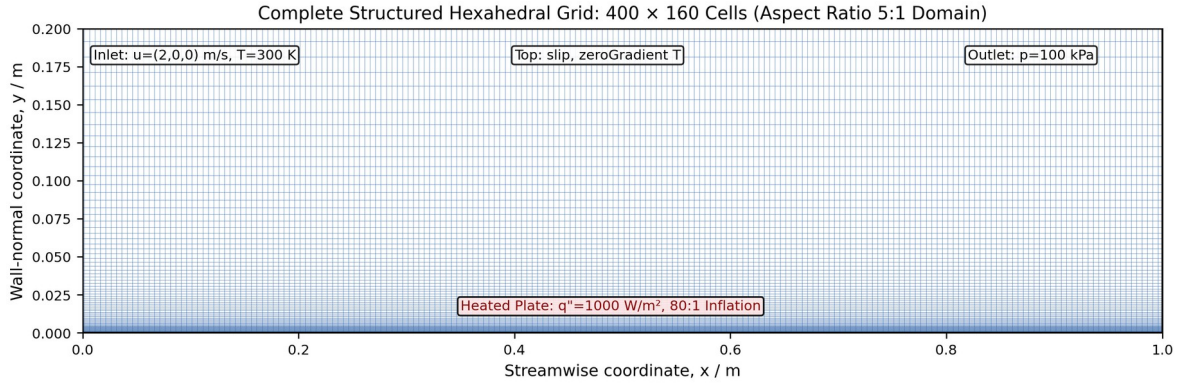


Fig. 4: Boundary patch locations on computational domain.

Configured OpenFOAM Patches:

- Inlet imposes uniform 2 m/s velocity and 300 K.
- Heated plate enforces no-slip and 1000 W/m² heater flux.
- Top boundary enforces slip velocity and zero thermal gradient.

Outlet: 100 kPa reference pressure with inletOutlet outflow condition.
Sides: Empty patch type enforcing two-dimensional planar behavior.

Surface Radiation Model

Wall Energy Balance (First-Law Closure):

$$q''_{heater} = q''_{conv} + q''_{rad}$$

Stefan-Boltzmann Radiation to Enclosure:

$$q''_{rad} = \varepsilon \sigma (T_w^4 - T_{sur}^4)$$

Objective Function Solved for Tw:

$$f(T_w) = -k \left(\frac{dT}{dy}\right)_w + \varepsilon \sigma (T_w^4 - T_{sur}^4) - q''_{heater} = 0$$

Newton-Raphson Wall Solver:

- Uses codedFixedValue boundary condition with Newton-Raphson iteration.
- Emissivity equals 0.0 for baseline and 0.8 for radiation.
- Models radiation to 300 K surroundings without gas absorption.

Stefan-Boltzmann constant sigma = 5.670374 x 10^-8 W/(m² K⁴).
Surrounding enclosure is held fixed at T_sur = 300 K.

CFD Solver and Numerical Setup

Setting / Parameter	Implementation	Details / Configured Values
CFD Framework	OpenFOAM Foundation v13	Standard finite volume formulation
Mesh Generator	blockMesh	Parametric structured hexahedral generation
Solver Application	foamRun	Incompressible steady/transient PIMPLE algorithm
Divergence Schemes	linearUpwind Gauss	Second-order upwind for div(phi,U) and div(phi,T)
Laplacian Schemes	Gauss linear orthogonal	Second-order central differencing with orthogonal correction
Pressure Solver	GAMG (DIC smoother)	Tolerance: 10^-7, relative tolerance: 0.01
Momentum & Energy Solver	PBiCGStab (DILU)	Tolerance: 10^-8, relative tolerance: 0.01
Run Controls	1500 iterations	deltaT = 1, final time index = 1500

- PIMPLE algorithm integrates pressure-velocity-energy coupling across 1500 iterations.
- Linear equation residuals drop below 10^-7 for all solved variables.
- Low linear solver residuals alone do not prove numerical convergence.

Mesh and Grid Sensitivity

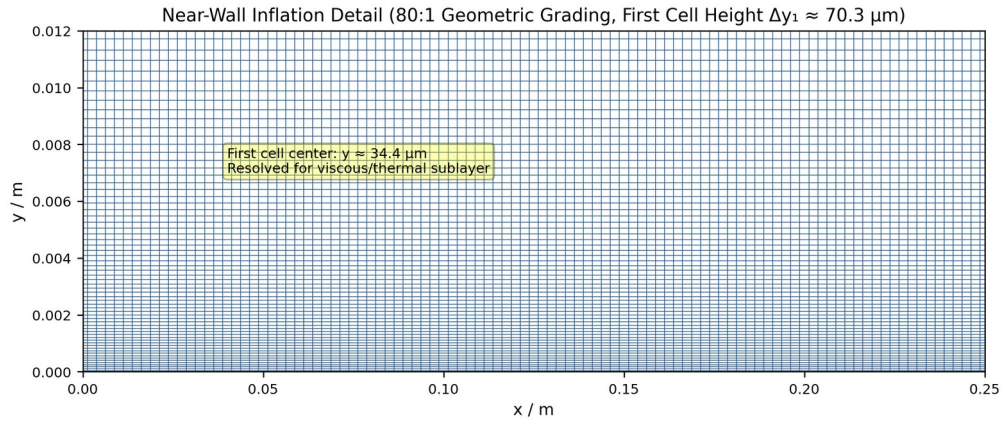


Fig. 5: Near-wall mesh zoom showing 80:1 wall grading.

Quantity	Coarse Mesh	Fine Mesh
Cells	16,000	64,000
Grid dimensions	200 x 80	400 x 160
First cell height	142.5 μm	70.3 μm
Maximum aspect ratio	36.63	36.34

Two meshes have been tested. Grid independence has not been demonstrated.

- Without radiation, trailing-edge wall temperature drops 51.22 K on fine mesh.
- With radiation, trailing-edge wall temperature drops 27.68 K on fine mesh.
- A third systematically refined mesh and GCI study are required.

Radiation Versus No-Radiation Results

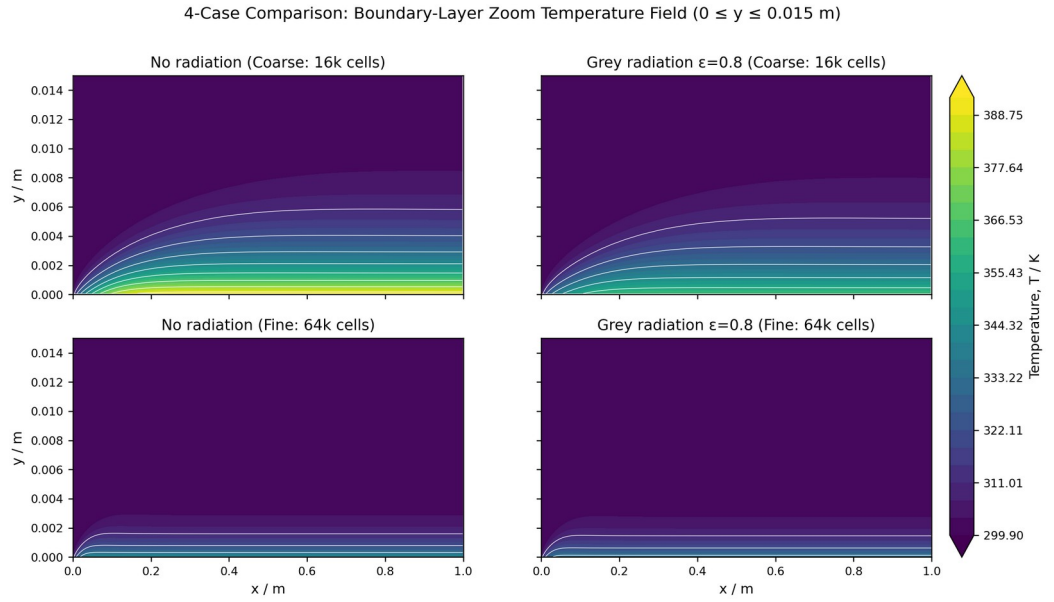


Fig. 6: 2 x 2 thermal boundary layer comparison.

Case	Reported Tw (x = 1.0 m)
No radiation, coarse	394.90 K
Radiation, coarse	364.38 K
No radiation, fine	343.68 K
Radiation, fine	336.70 K

- Radiation lowers reported wall temperature across both tested mesh levels.
- Temperature reduction is 30.52 K on coarse and 6.98 K on fine.
- Large grid difference prevents treating either reduction as mesh-independent.

Heat-Flux Partition

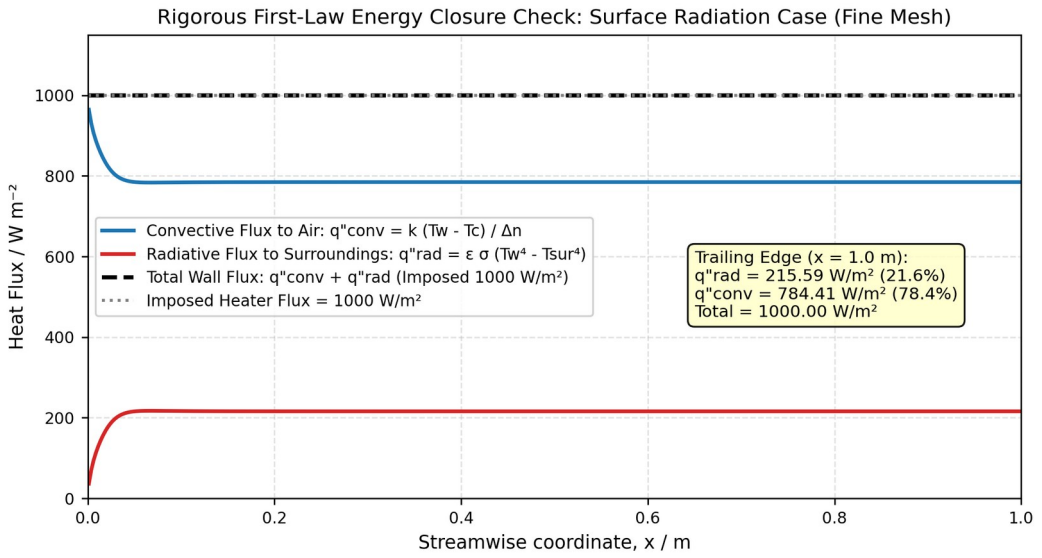


Fig. 7: Convective and radiative flux partition along plate.

Quantity (at x = 1.0 m)	Coarse, eps=0.8	Fine, eps=0.8
Wall temperature	364.38 K	336.70 K
Radiative heat flux	432.21 W/m²	215.59 W/m²
Convective heat flux	567.79 W/m²	784.41 W/m²
Radiative fraction	43.22%	21.56%

- Radiative fraction is defined as f_rad = (q''_rad / q''_heater) * 100%.
- Reported radiation fraction depends strongly on grid in current results.
- Flux sum balance is satisfied by construction, not independent proof.

Velocity Field and Momentum Coupling

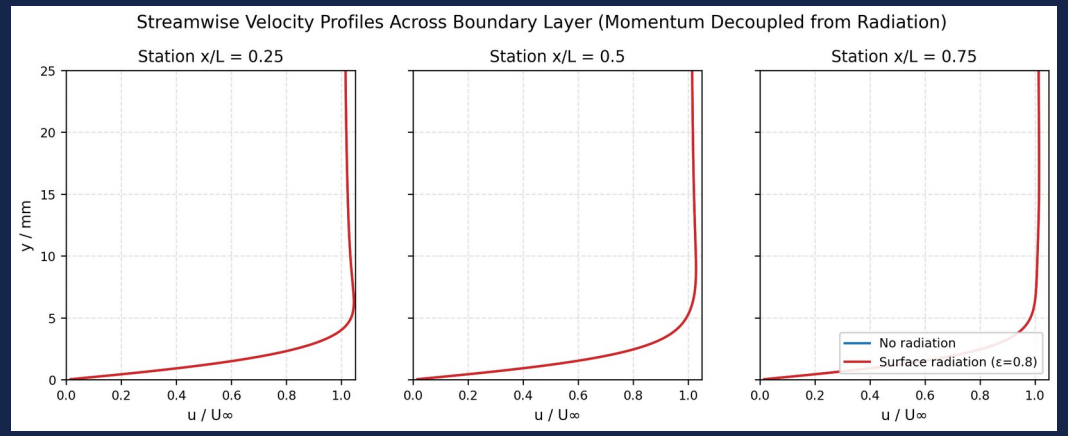


Fig. 8: Streamwise velocity profiles across stations.

< 10^-12 m/s

Maximum Velocity Field Difference |u_rad - u_norad|

- Constant density and viscosity decouple momentum from the thermal energy equation.
- Buoyancy is disabled, leaving momentum equations completely independent of temperature.
- Radiation changes thermal boundary layer without altering velocity solution.

Analytical Boundary-Layer Comparison

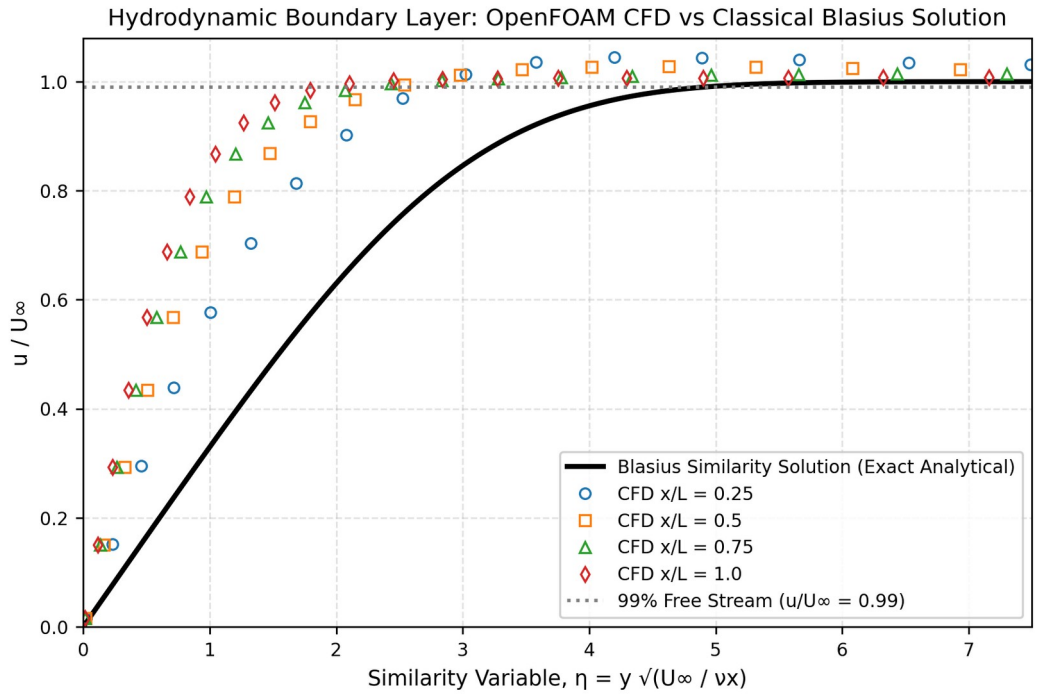


Fig. 9: Similarity profile comparison (eta vs u/U_inf).

5.38 mm vs 14.1 mm

Boundary-layer thickness delta_99 at x = 1.0 m (CFD reports 5.38 mm; Blasius predicts 14.1 mm)

875 vs 143

Trailing-edge Nusselt number Nu_L (CFD reports 875; laminar correlation predicts ~143)

- Similarity variable defined as eta = y * sqrt(U_inf / (nu x)).
- Discrepancies arise from domain height confinement and mesh resolution.
- Results are marked as provisional pending resolution of discrepancies.

Numerical Audit and Limitations

Verified from Current Files:

✓ Solver completed 1500 iterations with linear residuals below 10^-7.

✓ Newton-Raphson scheme converges non-linear wall energy equation.

✓ Momentum field decouples strictly under constant fluid properties.

Requires Further Verification:

! Grid independence: 51.22 K wall temperature shift between meshes.

! Discrepancy between 5.38 mm CFD thickness and 14.1 mm Blasius.

! Wall heat flux needs independent calculation from thermal gradient.

Classification Note: Unresolved numerical values are marked as provisional. Iterative convergence does not equal grid independence or experimental validation.

Conclusions and Next Work

1. Resolve Discrepancies
Investigate thickness (5.38 vs 14.1 mm) and Nusselt number (875 vs 143).

2. Independent Heat Flux
Calculate wall flux from -k(dT/dy)_w and check global energy balance.

3. Third Refined Mesh
Execute an 800 x 320 structured grid to establish asymptotic convergence.

4. Formal GCI Study
Compute ASME VV 20 Grid Convergence Index error bands for Tw and Nu.

5. Recheck Similarity
Match domain height and inlet conditions to semi-infinite assumptions.

6. Parameter Sensitivity
Vary emissivity (0 to 1) and flux (500 to 2000 W/m²) after verification.

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Conclusion: Radiation methodology is established, but validation requires resolving grid sensitivity and analytical discrepancies.